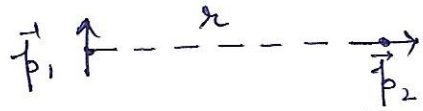
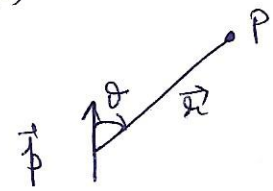
1.



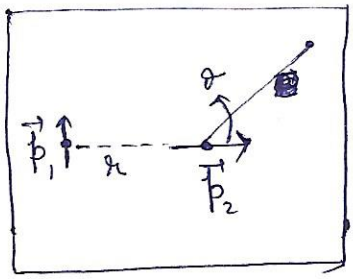
<u>To calculate torque on $\vec{p}_1$ due to $\vec{p}_2$:</u>

Field due to a dipole aligned along z-axis is given by

$$E = \frac{p}{4\pi\epsilon_0 r^3}\left[2\cos\theta\,(\hat{r}) + \sin\theta\,(\hat{\theta})\right] \quad \text{—(i)}$$



Therefore $\vec{E}_{\vec{p}_2}$ (at pos. of $\vec{p}_1$) =



$$= \frac{2p_2\cos\pi}{4\pi\epsilon_0 r^3}(\hat{r}) + \frac{p_2\sin\pi}{4\pi\epsilon_0 r^3}\hat{\theta}$$

$$= \frac{2p_2}{4\pi\epsilon_0 r^3}(-\hat{r})$$ (pointing along right)

Torque at $\vec{p}_1$, $\vec{N}_{p_1} = \vec{p}_1 \times \vec{E}_{p_2} = \dfrac{2p_1p_2}{4\pi\epsilon_0 r^3}$ (pointing into the page)

<u>To calculate torque on $\vec{p}_2$ due to $\vec{p}_1$:</u>

$\vec{E}_{\vec{p}_1}$ due to $\vec{p}_1$ (at pos. of $\vec{p}_2$) $= \dfrac{p_1(\sin\pi/2)\hat{\theta}}{4\pi\epsilon_0 r^3}$ [according to (i) with $\theta = \pi/2$]

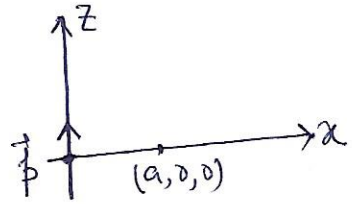
$$= \frac{p_1(\hat{\theta})}{4\pi\epsilon_0 r^3}$$ (pointing downwards at pos. of $\vec{p}_2$)

Torque $\vec{N}_{p_2}$ experienced by $\vec{p}_2$

$$= \vec{p}_2 \times \vec{E}_{p_1} = \frac{p_1p_2}{4\pi\epsilon_0 r^3}$$ (pointing into the page)

2. "Pure" dipole is situated at the origin pointing along z-direction.

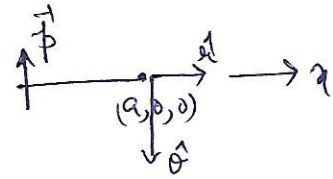
point charge placed at $(a,0,0)$



Electric field at $(a,0,0)$ due to $\vec{p}$ given by (i) with $\theta = \pi/2$

$\vec{E} = \dfrac{p}{4\pi\epsilon_0 r^3}\hat{\theta}$. But $\hat{\theta} = -\hat{z}$ at $(a,0,0)$

$\therefore \vec{E} = \dfrac{p}{4\pi\epsilon_0 r^3}(-\hat{z})$



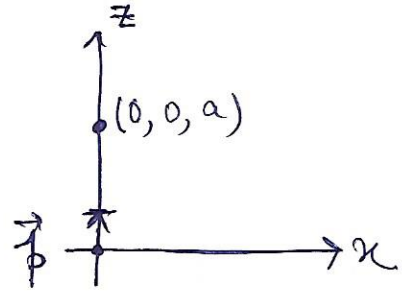
$= \dfrac{p}{\ }(-\hat{z})$

Thus at $(a, 0, 0)$, field is $\vec{E} = \dfrac{-p}{4\pi\epsilon_0 a^3}(\hat{z})$

Force on a charge particle is $q\vec{E}$

$$= \frac{-pq}{4\pi\epsilon_0 a^3}(\hat{z})$$

(b) Force on q at $(0, 0, a)$



$\vec{E}$ due to $\vec{p}$ at $(0, 0, a)$ given by

(i) with $\theta = 0$

$$\vec{E} = \frac{2p}{4\pi\epsilon_0 a^3}(\hat{r}) \text{ . Here } \hat{r} = \hat{z}$$

$$= \frac{2p}{4\pi\epsilon_0 a^3}(\hat{z}) \quad \therefore \boxed{\vec{F} = \frac{2pq}{4\pi\epsilon_0 a^3}\hat{z}}$$

(c) Work it takes to move q from $(a, 0, 0)$ to $(0, 0, a)$

$$= q\left[V(0, 0, a) - V(a, 0, 0)\right]$$

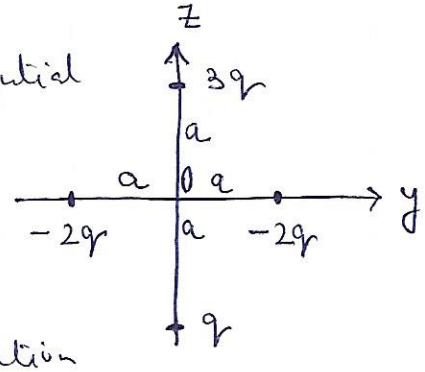
But $V = \dfrac{\vec{p}\cdot\hat{r}}{4\pi\epsilon_0 r^2}$ [potential at $\vec{r}$ due to $\vec{p}$ at origin]

$$= \frac{p\cos\theta}{4\pi\epsilon_0 r^2} \qquad V(0, 0, a) \longrightarrow \theta = 0$$

$$V(a, 0, 0) \longrightarrow \theta = \pi/2$$

$$W = q\left[\frac{p}{4\pi\epsilon_0 a^2} - 0\right] = \frac{pq}{4\pi\epsilon_0 a^2}$$

3. We have to find an approximate potential at points far away from the origin

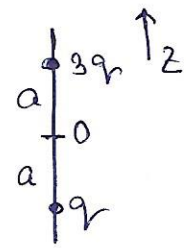


Clearly the total charge Q (which is also called the monopole moment) is $0\ (4q - 4q)$ and hence monopole contribution to the potential $\dfrac{Q}{4\pi\epsilon_0 r} = 0$ [here 'r' denotes distance from origin]

We have to calculate dipole moment of the system of charges. Whenever we have a collection of point charges, we know the dipole moment is given by

$$\vec{p} = \sum_i q_i \vec{r}_i' \qquad [\vec{r}_i' \rightarrow \text{distance of } i^{th} \text{ charge from origin}]$$

<u>For charges $+3q$ and q along Z-axis</u>

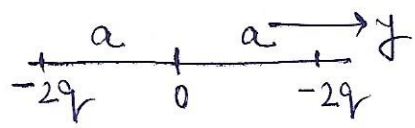


Choosing $\vec{r}_i' = +ve$ along $+ve$ Z-axis

$$\vec{p} = [3q \times a + q \times (-a)]\,\hat{z}$$

$$= 2qa\,(\hat{z})$$

<u>For charges $-2q$ and $-2q$ along y-axis</u>



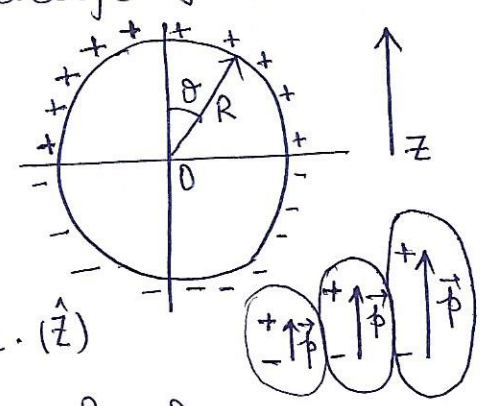
Choosing $\vec{r}_i' = +ve$ along $+ve$ y axis

$$\vec{p} = [(-2q \times (a)) + (-2q \times -a)]\,\hat{y}$$

$$= (-2qa + 2qa)\,\hat{y} = 0$$

$$V_{dip} \simeq \frac{(2qa)\cdot\hat{r}}{4\pi\epsilon_0 r^2} = \frac{2qa\cos\theta}{4\pi\epsilon_0 r^2} \qquad [\theta = \text{angle between Z-axis and distance } r \text{ from the origin}]$$

4. The spherical shell carries a surface charge $\sigma = k\cos\theta$. Due to this +ve and negative charges will be distributed on the surface as shown.



$$\vec{p} = q\,d\,(\hat{z}) = q\,z\,(\hat{z})$$

$$q = \int \rho\, d\tau = \oint \sigma\, da \quad \therefore \vec{p} = q\,z\,(\hat{z}) = \oint \sigma\, da\; z\cdot(\hat{z})$$

For carrying out the surface integral, put $z = R\cos\theta$

$$da = R^2 \sin\theta\, d\theta\, d\phi$$

$$p = \oint \sigma \, da \, z = \int_{\theta=0}^{\pi} \int_{\varphi=0}^{2\pi} k \cos\theta \cdot R^2 \sin\theta \, d\theta \, d\varphi \cdot R\cos\theta$$

$$= 2\pi k R^3 \underbrace{\int_0^{\pi} \cos^2\theta \sin\theta \, d\theta}_{2/3} = \frac{4}{3}\pi R^3 k$$

$$\therefore V_{dip} \simeq \frac{\frac{4}{3}\pi R^3 k \cos\theta}{4\pi\epsilon_0 r^2} = \frac{kR^3}{3\epsilon_0}\left(\frac{\cos\theta}{r^2}\right)$$

Actually this problem gives an <u>exact solution</u> when solving by Laplace's eqn $\nabla^2 V(r,\theta) = 0$ with an azimuthal symmetry (no φ- dependence) with the same solution as above.

$$V(r,\theta) = \sum_{l=0}^{\infty}\left(A_l r^l + \frac{B_l}{r^{l+1}}\right) P_l \cos\theta = \frac{kR^3}{3\epsilon_0}\left(\frac{\cos\theta}{r^2}\right)$$

Since this is an exact solution of above problem, it is clear that higher order multipoles in the approximate method (multipole expansion) are zero and only dipole term is present.